
Vehicle Market Analytics System

Entity Relationship Diagram & Relational Schema

Milestone 1 — Database Design Documentation

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Schema Overview

12	17	3NF	MySQL 8
Tables	Relationships	Normal Form	Engine

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1. Introduction

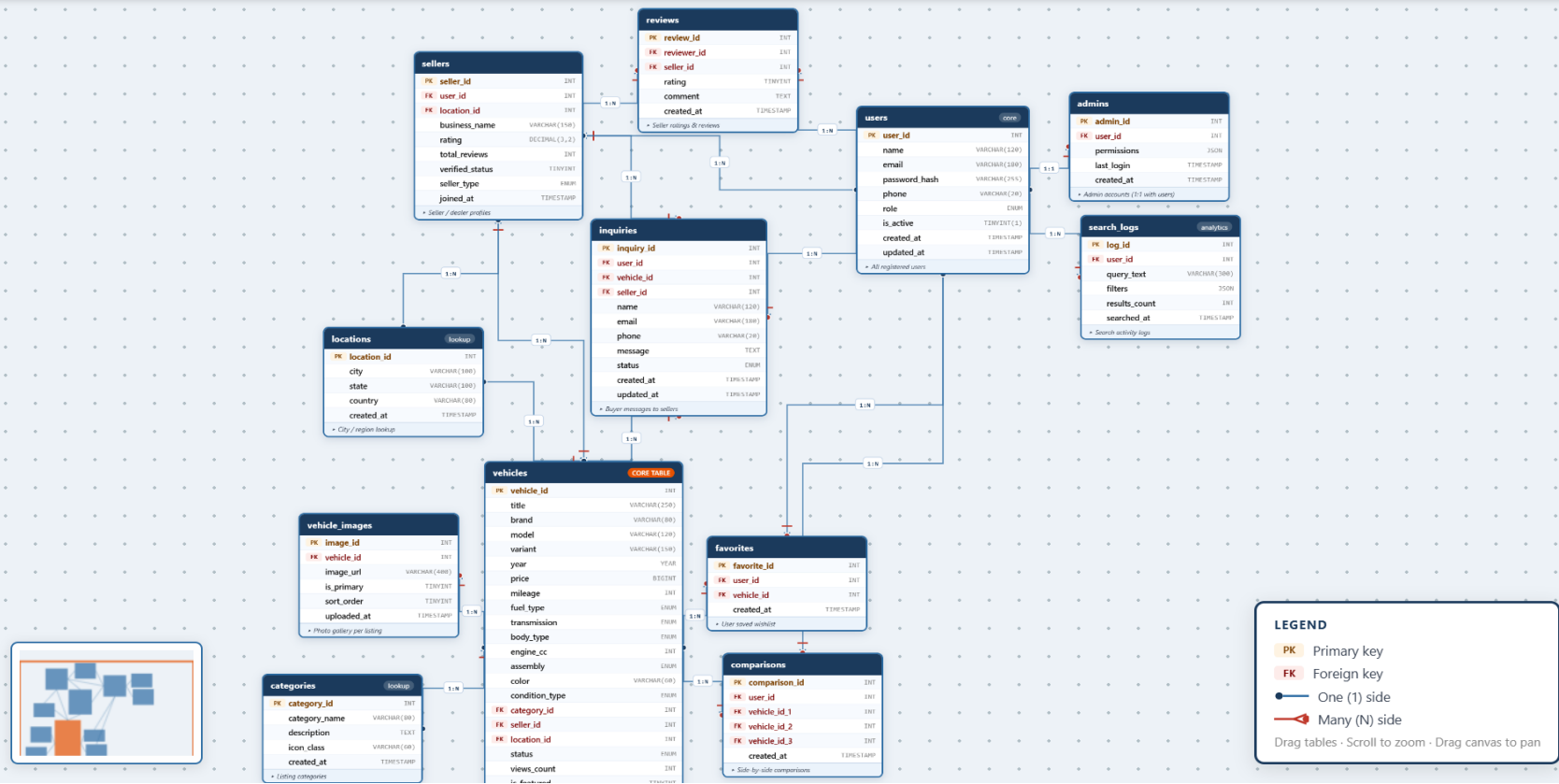
The **Vehicle Market Analytics System** aggregates and presents used vehicle listings from Pak-Wheels — Pakistan's largest automotive marketplace. The system is built on a **MySQL** relational database normalised to **3NF**, a Python **Flask** REST API, and a JavaScript frontend dashboard. This document covers Milestone 1: the complete Entity Relationship Diagram (ERD) for all 12 database tables and the relational schema for each table.

2. Entity Relationship Diagram

Figure 1 shows all 12 tables with their key columns. **PK** = Primary Key **FK** = Foreign Key
Arrows: child FK → parent PK.

PakWheels Vehicle Marketplace — Entity Relationship Diagram

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LEGEND

- PK Primary key
- FK Foreign key
- One (1) side
- Many (N) side

Drag tables · Scroll to zoom · Drag canvas to pan

3. Relational Database Schema

PK = Primary Key **FK** = Foreign Key **NN** = NOT NULL

3.1 locations

Geographic lookup table shared by sellers and vehicles.

Column	Type	Constraint	PK	FK	Notes
location_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
city	VARCHAR(100)	NOT NULL			City name.
state	VARCHAR(100)	NOT NULL			Province / region.
country	VARCHAR(80)	DEFAULT 'Pakistan'			Country name.

Relationships: sellers.location_id → locations vehicles.location_id → locations

3.2 categories

Lookup table for vehicle listing categories (Cars, Bikes, Vans).

Column	Type	Constraint	PK	FK	Notes
category_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
category_name	VARCHAR(80)	NOT NULL, UNIQUE			Category label.
description	TEXT	NULL			Optional description.

Relationships: vehicles.category_id → categories (SET NULL on delete)

3.3 users

Central account table for all registered users. role controls access level.

Column	Type	Constraint	PK	FK	Notes
user_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
name	VARCHAR(120)	NOT NULL			Full display name.
email	VARCHAR(180)	NOT NULL, UNIQUE			Login email.
password_hash	VARCHAR(255)	NOT NULL			Bcrypt hash.
phone	VARCHAR(20)	NULL			Contact number.
role	ENUM	DEFAULT 'buyer'			admin / seller / buyer.
is_active	TINYINT(1)	DEFAULT 1			Soft-delete flag.

Relationships: admins, sellers, favorites, comparisons, inquiries, reviews, search_logs → users

3.4 admins

Extends users for administrator accounts. One-to-one relationship.

Column	Type	Constraint	PK	FK	Notes
admin_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
user_id	INT UNSIGNED	NOT NULL, UNIQUE		•	FK → users. CASCADE.
permissions	JSON	DEFAULT '{}'			Admin access rights.

Relationships: admins.user_id → users.user_id (1:1, CASCADE)

3.5 sellers

Seller profiles extending users. Stores dealer or individual information.

Column	Type	Constraint	PK	FK	Notes
seller_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
user_id	INT UNSIGNED	NOT NULL, UNIQUE		•	FK → users. CASCADE.
location_id	INT UNSIGNED	NULL		•	FK → locations. SET NULL.
business_name	VARCHAR(150)	NULL			Dealer / showroom name.
rating	DECIMAL(3,2)	DEFAULT 0.00			Aggregate rating (0–5).
seller_type	ENUM	DEFAULT 'individual'			individual/dealer/certified.

Relationships: vehicles.seller_id → sellers inquiries.seller_id → sellers reviews.seller_id → sellers

3.6 vehicles (Core Table)

Primary entity. Every vehicle listing is stored here; references sellers, locations, and categories.

Column	Type	Constraint	PK	FK	Notes
vehicle_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
brand	VARCHAR(80)	NOT NULL			Manufacturer name.
model	VARCHAR(120)	NOT NULL			Model name.
variant	VARCHAR(150)	NULL			Trim / variant.
year	YEAR	CHECK > 1980			Manufacturing year.
price	BIGINT UNSIGNED	CHECK > 0			Listed price (PKR).
mileage	INT UNSIGNED	DEFAULT 0			Odometer (km).
fuel_type	ENUM	NOT NULL			Petrol/Diesel/Hybrid/EV/CNG.
transmission	ENUM	NOT NULL			Automatic/Manual/CVT/DCT.
body_type	ENUM	NOT NULL			Sedan/Hatchback/SUV/etc.
engine_cc	INT UNSIGNED	NULL			Displacement (cc).
category_id	INT UNSIGNED	NULL		•	FK → categories. SET NULL.
seller_id	INT UNSIGNED	NOT NULL		•	FK → sellers. CASCADE.
location_id	INT UNSIGNED	NULL		•	FK → locations. SET NULL.
status	ENUM	DEFAULT 'available'			available/sold/featured.

Relationships: *vehicle_images, favorites, comparisons, inquiries* → *vehicles*

3.7 vehicle_images

Stores image URLs per vehicle. *is_primary* marks the cover image.

Column	Type	Constraint	PK	FK	Notes
image_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
vehicle_id	INT UNSIGNED	NOT NULL		•	FK → vehicles. CASCADE.
image_url	VARCHAR(400)	NOT NULL			CDN image URL.
is_primary	TINYINT(1)	DEFAULT 0			Cover image flag.

Relationships: *vehicle_images.vehicle_id* → *vehicles* (CASCADE)

3.8 favorites

Junction table: vehicles saved to a user's wishlist. Unique (*user_id, vehicle_id*) enforced.

Column	Type	Constraint	PK	FK	Notes
favorite_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
user_id	INT UNSIGNED	NOT NULL		•	FK → users. CASCADE.
vehicle_id	INT UNSIGNED	NOT NULL		•	FK → vehicles. CASCADE.

Relationships: *UNIQUE (user_id, vehicle_id)*

3.9 comparisons

Side-by-side vehicle comparisons. Up to three vehicles; third slot is optional.

Column	Type	Constraint	PK	FK	Notes
comparison_	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
user_id	INT UNSIGNED	NOT NULL		•	FK → users. CASCADE.
vehicle_id_1	INT UNSIGNED	NOT NULL		•	First vehicle.
vehicle_id_2	INT UNSIGNED	NOT NULL		•	Second vehicle.
vehicle_id_3	INT UNSIGNED	NULL		•	Third (optional). SET NULL.

Relationships: *user_id → users vehicle_id_1/2/3 → vehicles*

3.10 inquiries

Buyer messages to sellers. Supports registered users and guests (*user_id* nullable).

Column	Type	Constraint	PK	FK	Notes
inquiry_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
user_id	INT UNSIGNED	NULL		•	FK → users. SET NULL.
vehicle_id	INT UNSIGNED	NOT NULL		•	FK → vehicles. CASCADE.
seller_id	INT UNSIGNED	NOT NULL		•	FK → sellers. CASCADE.
message	TEXT	NOT NULL			Inquiry message body.
status	ENUM	DEFAULT 'new'			new/read/replied/closed.

Relationships: *user_id → users (SET NULL) vehicle_id → vehicles seller_id → sellers*

3.11 reviews

Buyer ratings and reviews for sellers. Rating validated 1–5 via CHECK.

Column	Type	Constraint	PK	FK	Notes
review_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
reviewer_id	INT UNSIGNED	NOT NULL		•	FK → users. CASCADE.
seller_id	INT UNSIGNED	NOT NULL		•	FK → sellers. CASCADE.
rating	TINYINT UNSGND	CHECK (1-5)			Star rating.
comment	TEXT	NULL			Review text.

Relationships: reviewer_id → users seller_id → sellers

3.12 search_logs

Logs every search action for analytics and dashboard trends.

Column	Type	Constraint	PK	FK	Notes
log_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
user_id	INT UNSIGNED	NULL		•	FK → users. No cascade.
query_text	VARCHAR(300)	NULL			Search keywords.
filters	JSON	NULL			Serialised active filters.
results_cou	INT UNSIGNED	NULL			Results returned.

Relationships: search_logs.user_id → users (analytics rows retained on delete)

4. Relationships Summary

Parent	Child	Type	On Delete	Description
users	admins	1:1	CASCADE	Each admin maps to one user.
users	sellers	1:1	CASCADE	Each seller belongs to one user.
locations	sellers	1:N	SET NULL	One city, many sellers.
locations	vehicles	1:N	SET NULL	One city, many listings.
categories	vehicles	1:N	SET NULL	One category, many listings.
sellers	vehicles	1:N	CASCADE	One seller, many listings.
vehicles	vehicle_imag	1:N	CASCADE	One listing, many images.
users	favorites	1:N	CASCADE	One user, many saved vehicles.
vehicles	favorites	1:N	CASCADE	One vehicle, saved by many users.
users	comparisons	1:N	CASCADE	One user, many comparisons.
vehicles	comparisons	1:N	CASCADE/	Vehicle in multiple comparisons.
users	inquiries	1:N	SET NULL	Guest support; user_id nullable.
vehicles	inquiries	1:N	CASCADE	One listing, many inquiries.
sellers	inquiries	1:N	CASCADE	One seller, many inquiries.
users	reviews	1:N	CASCADE	One user, many reviews.
sellers	reviews	1:N	CASCADE	One seller, many reviews.
users	search_logs	1:N	None	Analytics rows retained.